

SMe Note

Acknowledge: Sample Space: Ω Random Variables: $\mathbf{Y} = (Y_1, \dots, Y_n)$ Sample Data: $\mathbf{y} = (y_1, \dots, y_n)$ $f_{\mathbf{Y}}(\mathbf{y}; \theta) = \prod_i f_{Y_i}(y_i; \theta)$, if y_i are independent Distribution Function: $F_{\mathbf{Y}}(\mathbf{y}) = \mathbb{P}(Y_1 \leq y_1, \dots, Y_n \leq y_n)$
Parameter Space: Θ CDF: $F_Y(y)$ PDF/PMF: $f_Y(y)$ or $p_Y(y)$

1 Basic Knowledge

Def of expectation (Moments): For any well-behaved function h on $\mathbb{R}$. $\mathbb{E}[h(Y)] := \sum_i h(y_i) \mathbb{P}(Y = y_i)$ $\mathbb{E}[h(Y)] := \int_{\mathbb{R}} h(y) f_Y(y) dy$

Def of Model: A collection of distributions indexed by a parameter $\{F_{\mathbf{Y}}(\cdot|\theta); \theta \in \Theta\}$ is model.

Likelihood Function: $L(\theta; \mathbf{y}) = f_{\mathbf{Y}}(\mathbf{y}; \theta)$ **MLE:** $\hat{\theta}(\mathbf{y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{y})] = [U(\hat{\theta}) = 0]$ ps: MLE has invariant property

(Strong) Likelihood Principle: $L(\theta; x) \propto L(\theta; y)$ [Strong: $L_f(\theta; x) \propto L_g(\theta; z)$; f, g are statistical model] $\Rightarrow$ Same Inferential Conclusions.

Def of Identifiability: A model is identifiable if $f(y; \theta_1) = f(y; \theta_2), \forall y$ then $\theta_1 = \theta_2$ (i.e. $\text{map} : \theta \rightarrow f(y; \theta)$ is 1-1)

· Method: 1.By def $\Rightarrow$ ratio not depend on $y \Rightarrow \theta_1, \theta_2$ relationship. (with any additional constraint) 2.MLE not unique $\Rightarrow$ not identifiable 3.Give an example θ 4.Fisher information

Linear Algebra Knowledge: If A is symmetric matrix, with eigenvalues λ_n then $A + aI$ have eigenvalues $\lambda_n + a$. $\| \det(M) = \prod \lambda_n$ eigenvalues of $X^T X \geq 0$.

2 Properties of the Likelihood and the MLE

Positive [Semi-]definite matrix M: $\forall \mathbf{z} \in \mathbb{R}^n \setminus \mathbf{0}, \mathbf{z}^T \mathbf{M} \mathbf{z} > 0 [\geq 0]$ $\Leftrightarrow (> 0)^1$ n positive eigenvalues. 2 full rank. 3 invertible

Score Func / Vector U	Single: $U(\theta, \mathbf{Y}) = \ell'(\theta; \mathbf{Y})$	$\mathbb{E}[U] = 0$	$Var[U] = I(\theta)$
	Multiple: $\mathbf{U}(\theta) = [\partial_{\theta_1} \ell(\theta; \mathbf{Y}), \dots, \partial_{\theta_p} \ell(\theta; \mathbf{Y})]^T \in \mathbb{R}^p$	$\mathbb{E}[\mathbf{U}] = \mathbf{0}$	$Var[\mathbf{U}] = I(\theta)$ $I_{j,k} = -Cov(\frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_j}, \frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_k}) = -\mathbb{E}[\frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_j} \frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_k}]$
CLT of U	Single: $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, I(\theta_0)), n \rightarrow \infty$		
	Multiple: $\mathbf{U}(\theta_0; \mathbf{Y}) \sim \mathcal{N}_p(\mathbf{0}, I(\theta_0)), n \rightarrow \infty$	Also: $\mathbf{U}(\theta_0; \mathbf{Y})^T I^{-1}(\theta_0) \mathbf{U}(\theta_0; \mathbf{Y}) \sim \chi_p^2$	
Fisher Information I	Single: Observed: $J(\theta, \mathbf{y}) = -U'(\theta, \mathbf{y}) = -\ell''(\theta; \mathbf{y})$	Expected: $I(\theta) = -\mathbb{E}[\ell''(\theta; \mathbf{Y})]$ $\ $ Unit: $i(\theta) = I(\theta)/n = -\mathbb{E}[\ell''(\theta; Y_i)]$	
	Multiple: Observed: $\mathbf{J}(\theta; \mathbf{y}) \in \mathbb{R}^{p \times p}, J_{j,k} = -\frac{\partial^2 \ell(\theta, \mathbf{y})}{\partial \theta_j \partial \theta_k}$	Expected: $I(\theta) \in \mathbb{R}^{p \times p}, I_{j,k} = -\mathbb{E}[\frac{\partial^2 \ell(\theta, \mathbf{Y})}{\partial \theta_j \partial \theta_k}]$ ps: $I(\theta)$ always positive semi-definite	
$I \uparrow \Rightarrow \theta$ 对 θ 越敏感; 信息量大; 参数估计更精确		A model is <i>identifiable</i> iff $I(\theta)$ is positive definite. ps:(一维) identifiable if $I(\theta) \neq 0$	
CLT of MLE CLT of I	Single: $I(\theta_0) < \infty \Rightarrow \hat{\theta} \sim \mathcal{N}(\theta_0, I^{-1}(\theta_0)), n \rightarrow \infty$	$\widehat{Var}(\hat{\theta}) \approx I^{-1}(\hat{\theta})$ $ESE(\hat{\theta}) \approx \sqrt{I^{-1}(\hat{\theta})}$ estimated standard error	
	Multiple: $I(\theta_0) < \infty \Rightarrow \hat{\theta} \sim N_p(\theta_0, I^{-1}(\theta_0)), n \rightarrow \infty$	Also: $(\hat{\theta} - \theta_0)^T I(\theta_0) (\hat{\theta} - \theta_0) \sim \chi_p^2$	

2.1 For Single Parameter

*Random Variable / Function: $L(\theta; \mathbf{Y})$ is random function (NOT pdf). For fixed θ , it's r.v. $\hat{\theta}(\mathbf{Y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{Y})]$ is r.v. ps: I^{-1}, J^{-1} 是指其负一次方, 不是反函数 True Parameter: θ_0

Name	Test Statistic	Hypothesis	CI	Other
Score Test	Given $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, I(\theta_0))$ $T_{Score}(\mathbf{Y}) = \frac{U(\theta_0)}{\sqrt{I(\theta_0)}} \sim \mathcal{N}(0, 1)$	$H_0 : \theta = \theta_0$ 不用 MLE, 泰勒展开	$\left\{ \theta : \frac{U^2(\theta)}{I(\theta)} < \chi_{\alpha, 1}^2 \right\}$ or(approx): $\hat{\theta} \pm z_{\alpha/2} \sqrt{I^{-1}(\hat{\theta})}$	or $T_{Score}^2(\mathbf{Y}) \sim \chi_1^2$ $T_{Score}(\mathbf{Y}) \approx \frac{U(\theta_0)}{\sqrt{I(\hat{\theta})}}, \frac{U(\theta_0)}{\sqrt{I(\theta_0)}}, \frac{U(\theta_0)}{\sqrt{I(\hat{\theta})}}$
Wald Test or Maximum Likelihood(ML) Test	$T_{Wald}(\mathbf{Y}) = \sqrt{I(\theta_0)}(\hat{\theta} - \theta_0) = \frac{\hat{\theta} - \theta_0}{\sqrt{I^{-1}(\theta_0)}}$ $T_{Wald}(\mathbf{Y}) \sim \mathcal{N}(0, 1)$	$H_0 : \theta = \theta_0$ 构建 CI	$\left\{ \theta : (\hat{\theta} - \theta)^2 I(\theta) < \chi_{\alpha, 1}^2 \right\}$ or(approx): $\hat{\theta} \pm z_{\alpha/2} \sqrt{I^{-1}(\hat{\theta})}$	or $T_{Wald}^2 \sim \chi_1^2$; $n \rightarrow \infty, T_{Wald} \approx T_{Score}$ $T_{Wald}(\mathbf{Y}) \approx \frac{(\hat{\theta} - \theta_0)}{\sqrt{I^{-1}(\hat{\theta})}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{I^{-1}(\theta_0)}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{I^{-1}(\hat{\theta})}}$
log-Likelihood Ratio Test	$T_{LR}(\mathbf{Y}) = 2(\ell(\hat{\theta}) - \ell(\theta_0)) \sim \chi_1^2$ as $n \rightarrow \infty$ $LR = L(\theta_0)/L(\hat{\theta}), T_{LR} = -2\ln(LR)$	$H_0 : \theta = \theta_0$ 最小 CI, Most Powerful	$\left\{ \theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < \chi_{\alpha, 1}^2 \right\}$	ps: $2(\ell(\hat{\theta}) - \ell(\theta_0)) \approx (\theta_0 - \hat{\theta})^2 J(\hat{\theta})$
$T_{Score} \approx T_{Wald} \approx T_{LR}$ as $n \rightarrow \infty$; $J(\theta_0) \approx I(\theta_0)$ as $n \rightarrow \infty$; When $\ell(\theta)$ 是关于 θ 的二次函数 $\Rightarrow$ All same. Geometric: (T_S , 比斜率), (T_W , 比宽度), (T_{LR} , 比高度)				

2.2 For Single Parameter (Numerical Method)

Suppose initial value solution θ_0	Newton-Raphson Method ps: also called Newton's method.	Fisher's Method of Scoring ps: also called scoring method
Formula	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\ell''(\theta_r)} = \theta_r + \frac{U(\theta_r)}{J(\theta_r)}$	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\mathbb{E}[\ell''(\theta_r)]} = \theta_r + \frac{U(\theta_r)}{I(\theta_r)}$
Advantages	Converge <i>very quickly</i> near optimum	I is constant / easier to calculate; Converge for wider θ_0
Stopping Rule	Given tolerance level ε : 1 Stop after r if $ U(\theta_r) \leq \varepsilon$ (Common) or 2 Stop after $ \theta_r - \theta_{r-1} \leq \varepsilon$	

2.3 For Multiple Parameters

*此节中, θ 代表向量, $\theta = (\theta_1, \dots, \theta_p)$ I^{-1} 指其逆矩阵 True Parameter: θ_0

Name	Test Statistic ($T_{Score}^2, T_{Wald}^2, T_{LR}$)	Hypothesis	(Confidence Region) CR
Score Test	$\mathbf{U}(\theta_0)^T I^{-1}(\theta_0) \mathbf{U}(\theta_0) > z_{\alpha, p}$	$\theta = \theta_0$	$\{ \theta : \mathbf{U}(\hat{\theta})^T I^{-1}(\theta) \mathbf{U}(\hat{\theta}) < z_{\alpha, p} \}$
Wald Test	$(\hat{\theta} - \theta_0)^T I(\theta_0) (\hat{\theta} - \theta_0) > z_{\alpha, p}$	$\theta = \theta_0$	$\{ \theta : (\hat{\theta} - \theta)^T I(\theta) (\hat{\theta} - \theta) < z_{\alpha, p} \}$
log-Likelihood Ratio Test	$2(\ell(\hat{\theta}) - \ell(\theta_0)) > z_{\alpha, p}$	$\theta = \theta_0$	$\{ \theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha, p} \}$
$I(\theta_0) \approx I(\hat{\theta}), J(\theta_0), J(\hat{\theta})$ $ps : z_{\alpha, p} = \chi_{\alpha, p}^2$ T_{LR} : simple hypothesis testing. ($\dim(\mathcal{A}) = 0$) θ 都是向量			

Marginal Normal distribution $\hat{\theta}$:

$\hat{\theta}_j \sim \mathcal{N}(\theta_j, [I^{-1}(\underline{\theta}_0)]_{jj})$ θ_j : true para

$\cdot ESE(\hat{\theta}_j) = \sqrt{[I^{-1}(\hat{\theta})]_{jj}}$

$\cdot CI(\theta_j): \hat{\theta}_j \pm z_{\alpha} ESE(\hat{\theta}_j)$

Composite Hypothesis Test: $H_0 : \underline{\theta} \in \mathcal{A}, H_1 : \underline{\theta} \notin \mathcal{A}$ $\mathcal{A}$: reduced model $LR = \frac{\max_{\theta \in \mathcal{A}} L(\underline{\theta})}{\max_{\theta \in \Theta} L(\underline{\theta})} = \frac{L(\underline{\hat{\theta}}_{\mathcal{A}})}{L(\underline{\hat{\theta}}_{\Theta})}$ ps: 多维未知参数 H_0 只想其中几个时用 $\| \dim(\mathcal{A})$ 常数时为 0

· **Generalised Likelihood Ratio Test:** $T_{GLR} = -2\ln(LR) = 2(\ell(\underline{\hat{\theta}}) - \ell(\underline{\hat{\theta}}_{\mathcal{A}})) \sim \chi_{p-r}^2$ p : # of parameters. $r = \dim(\mathcal{A})$

2.4 For Multiple Parameter (Numerical Method)

Method (start at θ_0)	*此节中, θ 代表向量 Newton-Raphson Method	Fisher's Method of Scoring
Formula	$\theta_{r+1} = \theta_r - \mathbf{H}^{-1}(\theta_r) \mathbf{U}(\theta_r) = \theta_r + \mathbf{J}^{-1}(\theta_r) \mathbf{U}(\theta_r)$	$\theta_{r+1} = \theta_r - (\mathbb{E}[\mathbf{H}(\theta_r)])^{-1} \mathbf{U}(\theta_r) = \theta_r + I^{-1}(\theta_r) \mathbf{U}(\theta_r)$
ps: Jacobian / Hessian $\mathbf{H} = -\mathbf{J}$ $\ $ If $I = \text{diag}(a, b, \dots) \Rightarrow I^{-1} = \text{diag}(a^{-1}, b^{-1}, \dots)$		

3 Bayesian Inference

3.1 Bayesian Approach & Definitions

Basic Idea: Consider θ as a r.v.. Bayesian approach $\Rightarrow$ Probability are used to represent degree of believe about the unknown θ
Prior Distribution $p(\theta)$: Prior distribution $p(\theta)$ contains all available prior information about θ *before observing any data*.
· **Hyper-Parameters:** 由于 $p(\theta)$ 是已知, 服从某种分布 (with parameter γ). We denoted $p(\theta)$ as $p(\theta; \gamma)$ where γ is *Hyper-parameters*. 定义: $\gamma_{prior} \rightarrow \gamma_{post}$
Likelihood function $p(y|\theta) = f(y|\theta)$: Any observed data is considered as a realisation of r.v. Y with $f(y|\theta)$
Posterior Distribution $p(\theta|y)$: Posterior distribution $p(\theta|y)$, contains all available knowledge.
· **Bayes' Rule / Theorem:** $p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{f(y|\theta)p(\theta)}{m(y)}$ $p(y) = m(y) = \int_{\theta \in \Theta} f(y|\theta)p(\theta)d\theta$ (Normalizing constant)

Name	MAP _{Maximum A Posterior}	MMSE _{Posterior Mean / Bayesian Minimum Mean Squared Error Estimator}	Posterior Median $\hat{\theta}_{median}$
Single Para	$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta y)$	$\hat{\theta}_{MMSE} = \mathbb{E}[\theta y] = \int_{\theta \in \Theta} \theta p(\theta y)d\theta$ ps: 估算后验分布参数一般方法	$\frac{1}{2} = \int_{-\infty}^{\hat{\theta}_{median}} p(\theta y)d\theta = \int_{\hat{\theta}_{median}}^{\infty} p(\theta y)d\theta$
Name	Posterior Var	$C_{\alpha} = [a, b]$ $(1 - \alpha)100\%$ Bayesian Credible Intervals (or Simply Credible Intervals)	
Single Para	$Var[\theta y] = \int_{\theta \in \Theta} (\theta - \mathbb{E}[\theta y])^2 p(\theta y)d\theta$	$\mathbb{P}(a \leq \theta \leq b y) = \int_a^b p(\theta y)d\theta = 1 - \alpha$ ps: $\mathbb{P}(\theta < a y) = \mathbb{P}(\theta > b y) = \alpha/2$	

Bayesian Decision Theory: 目的: ¹Estimating parameter θ ²Test Hypothesis about θ ³Find CI. ps: 后两种有很多多种方法来判断

1. Choose a *Loss Function* $Q(\hat{\theta}, \theta)$ (损失函数, 衡量估计值与真实值的差距; 贝叶斯决策的目的是最小化损失函数的期望)
2. Minimises Posterior Expected Loss Function: $\hat{\theta}_Q(y) = \arg \min_{\hat{\theta}} \mathbb{E}[Q(\hat{\theta}, \theta)|y] = \arg \min_{\hat{\theta}} \int_{\theta \in \Theta} Q(\hat{\theta}, \theta)p(\theta|y)d\theta$

3.2 Exact Bayesian Inference with Conjugate priors

Conjugate Prior: For θ , $p(\theta)$ is *Conjugate Prior* for $p(y|\theta) = f(y|\theta)$ iff for $p(\theta)$ 用 Bayes' Rule 得到的 $p(y|\theta)$ 都在同一类型的分布
Conjugate Family: The collection of all *Conjugate Prior*.
Choice of Prior: Observed data number n : ¹ n small $\Rightarrow$ Posterior shows a great dependency of the choice of prior.
² n middle $\Rightarrow$ posterior less affected by choice of prior. ³ n large $\Rightarrow$ choice of prior has limited effect on posterior. (Likelihood has a strong impact)

Comparing the frequentist and Bayesian approaches: ¹ $\hat{\theta}_{MMSE} \approx \hat{\theta}_{MAP}$ for most cases, especially n large. ² n large: $\hat{\theta}_{MAP} \approx \hat{\theta}_{MLE}$, $CI \approx C_{\alpha}$

3.3 Approximate Bayesian Inference

Intractable Posterior: Non-Conjugate model. (not analytic, no closed-form)
Monte Carlo Methods: Any func $g(\theta) || I_g = \int_{\theta \in \Theta} g(\theta)p(\theta|y)d\theta \approx \frac{1}{M} \sum_{i=1}^M g(\theta^{(i)}) \xrightarrow{a.s.} I_g$ $\theta^{(i)} \sim p(\theta|y), i = 1, ..., M$
1. **Importance Sampling (IS):** ¹ Simulating samples from another distribution $q(\theta)$ (*Proposal Distribution*) Consistent, Biased for finite
 $^2 w^{(i)} = \frac{\pi(\theta^{(i)}|y)}{q(\theta^{(i)})}$, $\pi(\theta; y) = f(y|\theta)p(\theta)$ ³ IS estimator: $\hat{I}_g^M = \sum_{i=1}^M g(\theta^{(i)}) \bar{w}^{(i)} \xrightarrow{a.s.} I_g$, Normalized Weight: $\bar{w}^{(i)} = \frac{w^{(i)}}{\sum_{j=1}^M w^{(j)}}$ $\theta^{(i)} \sim q(\theta)$ ps: $\sum_{j=1}^M \bar{w}^{(j)} = 1$
Remark: Choice of $q(\theta)$ is important. $|| \hat{I}_g^M$ depends on: ¹# of M ; ²mismatch between proposal and the integrand of the integral. Good if $q(\theta) \propto g(\theta)p(\theta|y)$, difficult to achieve.: (

2. **Markov Chain Monte Carlo (MCMC):** MCMC 算法通过 Markov Chain 生成样本, 使得样本接近服从后验分布. (计算量大)
Laplace Approximation: (最简单的估算 posterior 的方法) $p(\theta|y) \approx \mathcal{N}(\theta_{MAP}; H^{-1}(\theta_{MAP}))$ 类似 MLE 的 J $H(\theta) = -\frac{d^2 \ln(p(\theta|y))}{d\theta^2}$ (适合 $p(\theta|y)$ 接近正态分布的时适用)
Variational Inference (VI): ¹ Choose $q(\theta; \eta)$, η 是 q 的参数. ² 选择衡量 $p(\theta|y), q(\theta; \eta)$ 差异的方法 e.g. **KL divergence:** $KL(q(\theta; \eta)||p(\theta|y)) = \int_{\theta \in \Theta} q(\theta; \eta) \ln(\frac{q(\theta; \eta)}{p(\theta|y)}) d\theta$
³ 优化参数 (Find parameters to minimise the divergence): $\eta^* = \arg \min_{\eta \in A_{\eta}} KL(q(\theta; \eta)||p(\theta|y))$ ⁴ $p(\theta|y) \approx q(\theta; \eta^*)$

Remark: Both Laplace Approximation & Variation Inference approximate $p(\theta|y)$ by a parameter family of distributions. $||$ 拉普拉斯是正态分布家族, VI 是更普遍的分

4 Linear Model

4.1 Simple Linear Model ($y = \beta x + \varepsilon$)

Model: $Y_i = \mu_i + \varepsilon_i$, $\mu_i = \beta x_i$ $i = 1, 2, ..., n$ $||$ Suppose ¹ ε_i independent ² $\mathbb{E}[\varepsilon_i] = 0$ ³ $Var[\varepsilon_i] = \sigma^2$ $||$ Y_i response variable, x predictor variable.

Least Square (LS) Estimation	RSS _{Residual Sum of Squares} : $S(\beta) = \sum (y_i - \mu_i)^2 = \sum (y_i - x_i \beta)^2$	LS Method: $\hat{\beta}_i = \arg \min_{\beta} S(\beta)$ ps: 求偏导 = 0 $ \hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2}$
$\mathbb{E}, Var$ For β	$\mathbb{E}[\hat{\beta}] = \beta, \text{unbiased}$ $ Var[\hat{\beta}] = \sigma_{\hat{\beta}}^2 = (\sum x_i^2)^{-1} \sigma^2$ (σ^2 known)	$Var[\hat{\beta}] = \sigma_{\hat{\beta}}^2 = (\sum x_i^2)^{-1} \cdot \sigma^2$ $ \sigma^2 = \frac{1}{n-1} \sum (y - x_i \hat{\beta})^2$
H_0, H_1 Test $ $ CI	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim \mathcal{N}(0, 1)$ (σ^2 known)	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim t_{n-1}$ (σ^2 unknown)
Other	Fitted Values: $\hat{\mu}_i = x_i \hat{\beta}$ $ $ Residuals: $\hat{\varepsilon}_i = y_i - \hat{\mu}_i$	For CI / PI / H_0/H_1 Test: Assume $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$
PI	σ^2 know: $\hat{y}_{new} \pm z_{\alpha/2} \sigma \sqrt{1 + \frac{x_0^2}{S_{xx}}}$	σ^2 unknown: $\hat{y}_{new} \pm t_{\alpha/2, n-1} \hat{\sigma}_{\hat{\beta}} \sqrt{1 + \frac{x_0^2}{S_{xx}}}$

4.2 Simple Linear Model ($y = \alpha + \beta x + \varepsilon$)

Model: $Y_i = \mu_i + \varepsilon_i$, $\mu_i = \alpha + \beta x_i$ $i = 1, 2, ..., n$ $||$ Suppose ¹ ε_i independent ² $\mathbb{E}[\varepsilon_i] = 0$ ³ $Var[\varepsilon_i] = \sigma^2$

SL Estimate	$S_{xy} = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - \frac{1}{n} (\sum x_i)(\sum y_i)$	LS Method: $\hat{\beta} = \frac{S_{xy}}{S_{xx}} \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$	$\hat{\alpha}, \hat{\beta} = \arg \min_{\alpha, \beta} S(\alpha, \beta) S(\alpha, \beta) = \sum (y_i - \mu_i)^2$
$\mathbb{E}, Var$ For β	$\mathbb{E}[\hat{\beta}] = \beta, \text{unbiased}$ $ Var[\hat{\beta}] = \sigma_{\hat{\beta}}^2 = \frac{\sigma^2}{S_{xx}}$	$Var[\hat{\beta}] = \sigma_{\hat{\beta}}^2 = \frac{\sigma^2}{S_{xx}}$	$\sigma^2 = \frac{1}{n-2} \sum (y - \alpha - \beta x)^2 = \frac{1}{n-2} (S_{yy} - \frac{S_{xy}^2}{S_{xx}})$
$\mathbb{E}, Var$ For α	$\mathbb{E}[\hat{\alpha}] = \alpha, \text{unbiased}$ $ Var[\hat{\alpha}] = \sigma_{\hat{\alpha}}^2 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$	$Var[\hat{\alpha}] = \sigma_{\hat{\alpha}}^2 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$	$\sigma^2 = \frac{1}{n-2} \sum (y - \alpha - \beta x)^2 = \frac{1}{n-2} (S_{yy} - \frac{S_{xy}^2}{S_{xx}})$
H_0, H_1 Test $ $ CI	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim \mathcal{N}(0, 1)$ (σ^2 known)	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim t_{n-2}$ (σ^2 unknown)	$T = (\hat{\alpha} - \alpha_0) / Var(\hat{\alpha}) \sim t_{n-2}$ (σ^2 unknown)
(ctd.) Other	Fitted Values: $\hat{\mu}_i = \hat{\alpha} + \hat{\beta} x_i$ 残差: $\hat{\varepsilon}_i = y_i - \hat{\mu}_i$ $ \sum \hat{\varepsilon}_i = 0, \sum \hat{\varepsilon}_i x_i = 0$	For CI / PI / H_0/H_1 Assume $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$	$T = (\hat{\alpha} - \alpha_0) / Var(\hat{\alpha}) \sim \mathcal{N}(0, 1)$ (σ^2 known)
PI $Corr = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$	σ^2 know: $\hat{y}_{new} \pm z_{\alpha/2} \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$	σ^2 已知: $\hat{y}_{new} \pm t_{\alpha/2, n-2} \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$	σ^2 未知: $\hat{y}_{new} \pm t_{\alpha/2, n-2} \hat{\sigma}_y \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$

ps: $Cov(\hat{\alpha}, \hat{\beta}) = -\frac{\sigma^2 \bar{x}}{S_{xx}}$ or $\hat{\sigma}^2$ $Cov(\hat{\alpha}, \hat{\beta}) = 0 \Rightarrow$ independent. $|| S_{yy} = SS_{regression} + SS_{residuals}$ where $SS_{regression} = \sum (\hat{y}_i - \bar{y})^2$, $SS_{residuals} = \sum (y_i - \hat{y}_i)^2$

ps: Correlation: $Cor(\hat{\alpha}, \hat{\beta}) = \frac{Cov(\hat{\alpha}, \hat{\beta})}{\sqrt{Var(\hat{\alpha}) Var(\hat{\beta})}} = -\frac{\bar{x}}{\sqrt{\frac{1}{S_{xx}} + \bar{x}^2}}$ or 使用估计的 $\Rightarrow$ If = 0 $\Rightarrow$ independent.

4.3 General Linear Model

Lemma| $\mathbb{E}$, Var in Matrix: If r.v. $\mathbf{v}$, $\mathbf{A}$, $\mathbf{b}$ constant $\Rightarrow {}^1\mathbb{E}[\mathbf{Av} + \mathbf{b}] = \mathbf{AE}[\mathbf{v}] + \mathbf{b}$ ${}^2Var[\mathbf{Av} + \mathbf{b}] = \mathbf{AVar}[\mathbf{v}]\mathbf{A}^T$ || $\downarrow$ 注意: 对于 fitted residual 同方差成立; $\hat{\beta}$ 和拟合残差独立

Metric Predictor: 设计矩阵用连续观测值. || **Factor Predictor:** 离散数据用指示函数 (0/1) 填满, 每个"小组"对应一个指示函数.

Linear Model: $Y_i = \mu_i + \varepsilon_i$, $i = 1, 2, \dots, n$ $\mu = f(\beta, \mathbf{x})$ || $f(\beta, \mathbf{x})$ linear. || $\varepsilon_i \stackrel{i.i.d.}{\sim}$ distribution. $\mathbb{E}[\varepsilon_i] = 0$ $Var[\varepsilon_i] = \sigma^2$

· **Matrix Expression:** $\mathbf{y} = \mathbf{X}\underline{\beta} + \underline{\varepsilon}$ || Response Vector: $\mathbf{y}$ | Design / Model Matrix: $\mathbf{X}$ | Parameters / Coefficients: $\underline{\beta}$ | Additive Noise Vector: $\underline{\varepsilon}$

· **Parameter Estimates:** $\hat{\underline{\beta}}$ | **Estimated / Fitted Value Mean Vector:** $\hat{\underline{\mu}} = \mathbf{X}\hat{\underline{\beta}}$ | **Estimated / Fitted Residual Vector:** $\hat{\underline{\varepsilon}} = \mathbf{y} - \hat{\underline{\mu}}$

4.4 Linear Model Theory

Orthonormal Matrix $\mathbf{Q}$	Def: $\mathbf{Q}$:= columns and rows are orthonormal vectors.	$\mathbf{Q}^T = \mathbf{Q}^{-1}$ $\mathbf{QQ}^T = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}_n$ $\forall \mathbf{v}, \ \mathbf{v}\ ^2 = \ \mathbf{Qv}\ ^2 = \ \mathbf{Q}^T\mathbf{v}\ ^2$
(ctd.) **	Matrix $\mathbf{X}$ Orthogonal iff $\mathbf{X}^T\mathbf{X}$ is diagonal matrix.	$\mathbf{X}^T\mathbf{X}$ always symmetric matrix $\Rightarrow \mathbf{V}_{\hat{\underline{\beta}}}$ is diagonal $\Rightarrow \beta_i$ independent
QR Decomposition	$\forall \mathbf{X} \in \mathbb{R}^{n \times p} \Rightarrow \mathbf{X} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} = \mathbf{QR}$ $\mathbf{R} \in \mathbb{R}^{p \times p}$ is upper triangular matrix; $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is orthonormal matrix; $\mathbf{Q}$ 前 p 列等于 $\mathbf{Q} \in \mathbb{R}^{n \times p}$	
Least Square (LS)	$RSS := S(\beta) = \sum_{i=1}^n \varepsilon_i^2(\beta) = \sum_{i=0}^n (y_i - \mu_i(\beta))^2 = \ \mathbf{y} - \underline{\mu}\ ^2 = \ \mathbf{y} - \mathbf{X}\underline{\beta}\ ^2 = \ \underline{\varepsilon}(\beta)\ ^2$	$\hat{\underline{\beta}} = \arg \min_{\beta} S(\beta) = \arg \min_{\beta} \ \mathbf{f} - \mathbf{R}\underline{\beta}\ ^2$
QR Method of LS	Let ${}^1\mathbf{Q}$ be orthonormal matrix ${}^2\mathbf{X} = \mathbf{QR}$ ${}^3\mathbf{Q}^T\mathbf{y} = \begin{pmatrix} \mathbf{f} \in \mathbb{R}^p \\ \mathbf{r} \in \mathbb{R}^{n-p} \end{pmatrix} \Rightarrow S(\beta) = \ \mathbf{Q}^T\underline{\varepsilon}(\beta)\ ^2 = \dots = \ \mathbf{f} - \mathbf{R}\underline{\beta}\ ^2 + \ \mathbf{r}\ ^2$	
QR Method (ctd.)	If $\mathbf{R}$ 满秩: $\hat{\underline{\beta}} = \mathbf{R}^{-1}\mathbf{f} = \mathbf{R}^{-1}\mathbf{Q}^T\mathbf{y}$ $RSS = \ \mathbf{r}\ ^2 = \ \underline{\varepsilon}(\hat{\underline{\beta}})\ ^2 = \ \mathbf{y} - \mathbf{X}\hat{\underline{\beta}}\ ^2$ $RSE = \frac{RSS}{n-p} = \hat{\sigma}^2$ $\text{ps: } \mathbf{f} = \mathbf{Q}^T\mathbf{y}$ $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_p, \mathbf{QQ}^T \neq \mathbf{I}$	

More Inference of LS:

Matrix Method of LS	$S(\beta) = (\mathbf{y} - \mathbf{X}\underline{\beta})^T(\mathbf{y} - \mathbf{X}\underline{\beta}) = \mathbf{y}^T\mathbf{y} - 2\mathbf{y}^T\mathbf{X}\underline{\beta} + \underline{\beta}^T\mathbf{X}^T\mathbf{X}\underline{\beta}$	$\hat{\underline{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$	$\mathbf{V}_{\hat{\underline{\beta}}} = Var[\hat{\underline{\beta}}] = (\mathbf{X}^T\mathbf{X})^{-1}\sigma^2$
Influence / Hat Matrix $\mathbf{A}$	$\mathbf{A} = \mathbf{QQ}^T = \mathbf{A}^2 = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ $\mathbf{A}$ is symmetric. # of (identifiable) parameters = $tr(\mathbf{A}) = tr(\mathbf{QQ}^T) = tr(\mathbf{I}_p) = p$		
$Var[\mathbf{y}], Var[\mathbf{f}], \hat{\underline{\mu}}, \hat{\underline{\beta}}$	$Var[\mathbf{y}] = Cov(\underline{\varepsilon}, \underline{\varepsilon}) = \sigma^2\mathbf{I}_n$ $Var[\mathbf{f}] = \mathbf{Q}^T\mathbf{Q}\sigma^2 = \mathbf{I}_p$ $\hat{\underline{\mu}} = \mathbf{X}\hat{\underline{\beta}} = \mathbf{Ay}$ $\hat{\underline{\beta}} = \mathbf{R}^{-1}\mathbf{Q}^T\mathbf{y}$		
$\mathbf{V}_{\hat{\underline{\beta}}}, \mathbf{V}_{\hat{\underline{\varepsilon}}}, \mathbf{V}_{\hat{\underline{\mu}}} \hat{\underline{\mu}}$ dist	$\mathbf{V}_{\hat{\underline{\beta}}} = Var[\hat{\underline{\beta}}] = \mathbf{R}^{-1}\mathbf{R}^{-T}\sigma^2$ $\mathbf{V}_{\hat{\underline{\varepsilon}}} = Var[\hat{\underline{\varepsilon}}] = (\mathbf{I}_n - \mathbf{A})\sigma^2$ $\mathbf{V}_{\hat{\underline{\mu}}} = Var[\hat{\underline{\mu}}] = \mathbf{A}\sigma^2$ $\hat{\underline{\mu}}$ is Degenerate Normal (退化正太分布, Σ 不满秩)		
$\mathbb{E}[\hat{\underline{\beta}}], \mathbb{E}[\hat{\underline{\varepsilon}}], \mathbb{E}[\hat{\underline{\mu}}] \hat{\underline{\varepsilon}}$ dist	$\mathbb{E}[\hat{\underline{\beta}}] = \underline{\beta}$ unbiased $\mathbb{E}[\hat{\underline{\varepsilon}}] = \mathbf{0}$ $\mathbb{E}[\hat{\underline{\mu}}] = \underline{\mu}$ (unbiased) $\hat{\underline{\varepsilon}}$ is Degenerate Normal (退化正太分布, Σ 不满秩)		
检查正交/线性无关: If $\mathbf{X}^T\mathbf{X}$ is diagonal $\Rightarrow \mathbf{X}$ Orthogonal. $\Rightarrow \mathbf{V}_{\hat{\underline{\beta}}}$ diagonal $\Rightarrow \beta_i$ independent.	Check Identifiable: 法 1: $det(\mathbf{X}^T\mathbf{X}) \neq 0$, identifiable. 法 2: $Rank(\mathbf{X}) = p$ 既列向量 independent.)		

Prediction | PI: ** Create a Prediction Matrix: $\mathbf{X}^*$ ($\mathbf{X}^*$ 形式和 $\mathbf{X}$ 保持一致, 数据变为按拟合后的每个点的数据) $\Rightarrow \sigma^2$ 已知 $\hat{\underline{\mu}}^* = \mathbf{X}^*\hat{\underline{\beta}}$, $\hat{\underline{\mu}}^* \sim \mathcal{N}(\underline{\mu}^*, \mathbf{X}^*(\mathbf{X}^T\mathbf{X})^{-1}(\mathbf{X}^*)^T\sigma^2)$

4.5 CI / Hypothesis Testing

Normality Assumption: Assume that $\underline{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2\mathbf{I}_n) \Rightarrow {}^1\mathbf{y} \sim \mathcal{N}_n(\mathbf{X}\underline{\beta}, \mathbf{I}_n\sigma^2)$ ${}^2\hat{\underline{\beta}} \sim \mathcal{N}_p(\underline{\beta}, \mathbf{V}_{\hat{\underline{\beta}}})$ || 对于 Normal, LS $\hat{\underline{\beta}} = \text{MLE } \hat{\underline{\beta}}$

$\mathbf{V}_{\mathbf{Q}^T\mathbf{y}}, \mathbb{E}[\mathbf{Q}^T\mathbf{y}] \hat{\sigma}^2, \hat{\mathbf{V}}_{\hat{\underline{\beta}}}$	$\mathbf{V}_{\mathbf{Q}^T\mathbf{y}} = Var[\mathbf{Q}^T\mathbf{y}] = \mathbf{I}_n\sigma^2$ $\mathbb{E}[\mathbf{Q}^T\mathbf{y}] = \mathbb{E}\left[\begin{pmatrix} \mathbf{f} \\ \mathbf{r} \end{pmatrix}\right] = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}\underline{\beta}$ $\hat{\sigma}^2 = \frac{\ \mathbf{r}\ ^2}{n-p}$ (unbiased) $\hat{\mathbf{V}}_{\hat{\underline{\beta}}} = \mathbf{R}^{-1}\mathbf{R}^{-T}\hat{\sigma}^2$ (unbiased) $\hat{\underline{\beta}}, \hat{\sigma}^2$ independent
$\mathbf{f}, \mathbf{r}$ Distribution	Known σ^2 : $\mathbf{f} \sim \mathcal{N}(\mathbf{R}\underline{\beta}, \mathbf{I}_p\sigma^2)$ $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{n-p}\sigma^2)$ Unknown σ^2 : $\hat{\sigma}_{\hat{\beta}_i} = \sigma_{\hat{\beta}_i}\hat{\sigma}/\sigma$ $\hat{\sigma}_{\hat{\beta}_i} = \sqrt{[\hat{\mathbf{V}}_{\hat{\underline{\beta}}}]_{ii}}$ $\sigma_{\hat{\beta}_i} = \sqrt{[\mathbf{V}_{\hat{\underline{\beta}}}]_{ii}}$
Test Statistic	For $\underline{\beta}$ 整体: $T_{known \sigma^2}(\mathbf{Y}) = \frac{1}{\hat{\sigma}^2}\ \mathbf{r}\ ^2 = \frac{1}{\hat{\sigma}^2}\sum_{i=1}^{n-p} r_i^2 \sim \chi^2_{n-p}$ For β_i 个体: $T_{unknown \sigma^2}(\mathbf{Y}) = (\hat{\beta}_i - \beta_i)/\hat{\sigma}_{\hat{\beta}_i} \sim t_{n-p}$

Testing Constraints in the Model: 如果我们只是想对一部分 β_i 进行假设检验 H_0, H_1 . 使用此方法.

1 Create Selection Matrix $\mathbf{C} \in \mathbb{R}^{q \times p}$ q 检验个数, p 参数总数 e.g. $\mathbf{C} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ 如果我们要检验 β_1, β_2 但总数有 3 个参数.	3 Under H_0 : $\mathbf{C}\hat{\underline{\beta}} - \mathbf{d} \sim \mathcal{N}(\mathbf{0}, \mathbf{CV}_{\hat{\underline{\beta}}} \mathbf{C}^T)$ $H_0 : \mathbf{C}\hat{\underline{\beta}} = \mathbf{d}$ vs $H_1 : \mathbf{C}\hat{\underline{\beta}} \neq \mathbf{d}$ $\mathbf{CV}_{\hat{\underline{\beta}}} \mathbf{C}^T = \mathbf{L}^T \mathbf{L}$ (Cholesky Decomposition) $\mathbf{L}^{-T}(\mathbf{C}\hat{\underline{\beta}} - \mathbf{d}) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_q)$ $\star RSS_{reduced} = S(\hat{\underline{\beta}}_{reduced})$ $RSS_{full} = S(\hat{\underline{\beta}}_{full})$	4 Test Statistic: $\text{ps: } T > F_{\alpha}$ 拒绝 (σ^2 known): $T(\mathbf{Y}) = (\mathbf{C}\hat{\underline{\beta}} - \mathbf{d})^T(\mathbf{CV}_{\hat{\underline{\beta}}} \mathbf{C}^T)^{-1}(\mathbf{C}\hat{\underline{\beta}} - \mathbf{d}) \sim \chi^2_q$ (σ^2 unknown): $T(\mathbf{Y}) = \frac{1}{q}(\mathbf{C}\hat{\underline{\beta}} - \mathbf{d})^T(\mathbf{C}\hat{\mathbf{V}}_{\hat{\underline{\beta}}} \mathbf{C}^T)^{-1}(\mathbf{C}\hat{\underline{\beta}} - \mathbf{d}) \sim F_{q, n-p}$ (if $\mathbf{d} = \mathbf{0}$) = $\frac{(RSS_{reduced} - RSS_{full})/q}{RSS_{full}/(n-p)} \sim F_{q, n-p}$
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4.6 Assumptions || Diagnostics

R^2, R^2_{adj} Metric: $R^2 = 1 - \frac{\frac{1}{n}RSS}{\frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\frac{1}{n}\sum_{i=1}^n \hat{\varepsilon}_i^2}{\frac{1}{n}\sum_{i=1}^n (y_i - \bar{y})^2}$ $RSS = S(\hat{\underline{\beta}}_{full})$ (接近 1, 拟合效果好) || $R^2_{adj} = 1 - \frac{\frac{1}{n-p}\sum_{i=1}^n \hat{\varepsilon}_i^2}{\frac{1}{n-1}\sum_{i=1}^n (y_i - \bar{y})^2}$ (avoid biased of R^2)

Graphical Assessment

- Residuals vs Fitted (Check linear assumption): $(\hat{\mu}_i, \hat{\varepsilon}_i)$
 1 平均值线在 0 附近, 越靠 0 越好. ($\Rightarrow \mathbb{E}[\hat{\varepsilon}_i] = 0$) || 2 有趋势/系统性 (systematic pattern in the mean of the residuals): violate the independence assumption.
 3 点在图上均匀分布 (spread / variability does not change much for different fitted values) $\rightarrow$ constant variance
- Scale-Location (Check constant variance assumption): $(\hat{y}_i, \sqrt{|e_i^*|}), e_i^* = \frac{e_i}{s_e}$ 越靠近 0.82 越好. (around a constant value)
- Normal QQ-Plot (Check Normal Assumption): $(z_i, e_i^*), z_i = \Phi^{-1}(\frac{i-0.5}{n})$
- Residuals vs Leverage (Check influential points): $(\mathbf{A}_{ii}, \hat{\varepsilon}_i/(\hat{\sigma}\sqrt{1 - \mathbf{A}_{ii}}))$ high leverage points: x 轴上远离大多数数据点. (对拟合结果影响大) 超过 cook's distance 认为影响大.
- 线性回归图像 (Check Independence assumption): $(\mathbf{x}, \mu_i)$ or (x_i, μ_i) (符合假设 $\rightarrow$ 随机在回归线上随机分布的; 如果存在聚集性点 (cluster points) $\Rightarrow$ 不符合 independence.)

Check balance of quantiles: 检查残差的分布是否对称. Method 1: $\frac{|\hat{\varepsilon}_{median}|}{\hat{\sigma}}$ Method 2: $\frac{|\hat{\varepsilon}_{Q3}| - |\hat{\varepsilon}_{Q1}|}{\hat{\sigma}}$ small $\Rightarrow$ no evidence against residuals not being symmetric

5 Distribution Table

Distribution	PDF	Mean	Var	Other
Bernoulli $X \sim Bern(p)$	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	p	$p(1-p)$	If X_i are i.i.d. $\sum X_i \sim Bin(n, p)$
Binomial $X \sim Bin(n, p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	$X, Y \sim Bin(n, p), Bin(m, p)$ $X + Y \sim Bin(n + m, p)$
For $X \sim Bin(n, p), n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda$, then $X \sim Poisson(\lambda)$		For $X \sim Bin(n, p), n \rightarrow \infty$ and pdf symmetric, $\Rightarrow X \sim N(np, np(1-p))$		
Geometric $X \sim Geom(p)$	$(1-p)^{x-1}p$	$1/p$	$(1-p)/p^2$	$\mathbb{E}[X^2] = 2/p^2$
Negative Binomial $X \sim NB(r, p)$	$\binom{r+x-1}{x}(1-p)^r p^x, r \geq 1$	$\frac{rp}{1-p}$	$\frac{rp}{(1-p)^2}$	或者 $p = 1 - q$, 注意!
Poisson $X \sim Po(\lambda)$	$e^{-\lambda} \cdot \frac{\lambda^x}{x!}$	λ	λ	$\mathbb{E}[X^2] = \lambda^2 + \lambda$
For $X, Y \sim Po(\lambda_1), Po(\lambda_2)$, then $X + Y \sim Po(\lambda_1 + \lambda_2)$				

Multinomial $X_1 + \dots + X_k = n$ $(X_1, \dots, X_k) \sim MN(n, p_1, \dots, p_k)$	$\frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}, 0 < p_i < 1$	np_i	$np_i(1 - p_i)$	
Uniform $X \sim \mathcal{U}([a, b])$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\mathbb{E}[X^2] = \frac{b^2+ab+a^2}{3}$
Normal $X \sim \mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\mathbf{Y} \sim N_k(\mu, \Sigma), \mu \text{ vector}$ $f_Y(y_1, \dots, y_k) = \frac{e^{-\frac{1}{2}(\mathbf{y}-\mu)^T \Sigma^{-1}(\mathbf{y}-\mu)}}{\sqrt{(2\pi)^k \det(\Sigma)}}$
For $X, Y \sim N(\mu_1, \sigma_1^2), Po(\mu_2, \sigma_2^2)$, then $aX \pm Y \pm c \sim Po(a\mu_1 \pm b\mu_2 \pm c, a^2\sigma_1^2 + b^2\sigma_2^2)$				
Exponential $X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\mathbb{E}[X^n] = \frac{n!}{\lambda^n}$
Beta $X \sim Beta(\alpha, \beta), \alpha, \beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, x \in [0, 1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
When $Beta(\alpha, \beta)$ has maximum probability, $x = (\alpha - 1)/(\alpha + \beta - 2)$				
Gamma $X \sim \Gamma(\alpha, \beta), \alpha, \beta > 0$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x \geq 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\Gamma(\alpha) = (\alpha - 1)!$ $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$
Student's t $X \sim t_v, v > 0$	$\frac{1}{\sqrt{v}B(\frac{v}{2}, \frac{1}{2})} (1 + \frac{x^2}{v})^{-\frac{v+1}{2}}$	0	$\frac{v}{v-2}$	If $Z \sim N(0, 1), Y \sim \chi_n^2$ $\frac{Z}{\sqrt{Y/n}} \sim t_n$
PDF Symmetric at zero and has more mass in the tails (compare to the standard normal distribution)				
When $n \rightarrow \infty$ it will converges to $N(0, 1)$. Cauchy Distribution: $X \sim t_1$ and its mean and var are undefined.				
Chi-Squared $X \sim \chi_v^2, v > 0$	$\frac{1}{2^{\frac{v}{2}} \Gamma(\frac{v}{2})} x^{\frac{v}{2}-1} e^{-\frac{x}{2}}, x \geq 0$	v	$2v$	-
If $X_i \sim N(0, 1)$, then $\sum X_i^2 \sim \chi_n^2$ If $X \sim \chi_n^2, Y \sim \chi_m^2$, then $X + Y \sim \chi_{n+m}^2$				
F $X \sim F(\lambda, v), v > 0$ ps: $f_{a,b;\alpha} = 1/f_{a,b;1-\alpha}$	$\frac{\lambda^{\frac{\lambda}{2}} v^{\frac{v}{2}}}{B(\frac{\lambda}{2}, \frac{v}{2})} x^{\frac{\lambda}{2}-1} (\lambda x + v)^{-\frac{\lambda+v}{2}}, x \geq 0$	$\frac{v}{v-2}$	$\frac{2v^2(\lambda+v-2)}{\lambda(v-2)^2(v-4)}, v > 4$	$Y_1, Y_2 \sim \chi_{n_1}^2, \chi_{n_2}^2, X = \frac{Y_1/n_1}{Y_2/n_2}$ $X \sim F_{n_1, n_2}, X^{-1} \sim F_{n_2, n_1}$

6 Review

6.1 Basic Concepts

Law of Total Probability: $\mathbb{P}(A) = \sum_n \mathbb{P}(A|B_n)\mathbb{P}(B_n)$, $\cup_n B_n = \Omega$, B_n disjoint

Type	Type I error	Type II error	-	H_0 is T	H_0 is F
Def	$\mathbb{P}(Rej H_0 H_0 \text{ is } T)$	$\mathbb{P}(Fail Rej H_0 H_0 \text{ is } F)$	Fail to Rej H_0	Right	Type II (β) 与 size 有关
			Rej H_0	Type I (α) 与 size 无关	Right (Power)

Significance Level: $\alpha = \mathbb{P}(Rej H_0|H_0 \text{ is } T)$

P-value: 1.One-side: $p = \mathbb{P}(T \geq t; T \sim H_0)$ 2.Two-sides: $p = 2 \min\{\mathbb{P}(T \geq t; T \sim H_0), \mathbb{P}(T \leq t; T \sim H_0)\}$

Power of Test: $\mathbb{P}(Rej H_0|H_0 \text{ is } F) = 1 - \beta = \mathbb{P}(Z \in C|\mu = \mu^*)$ · 1.Sample size ↑, Power ↑; 2. σ^2 ↓, Power ↑; 3.Effect size $|\theta_0 - \theta^*|$ ↑, Power ↑; 4. α ↑, Power ↑.

Def of Confidence Interval (CI): 100(1 - α)% Confidence Interval for θ is $[l(X), u(X)]$, $\mathbb{P}(l(X) < \theta < u(X)) = 1 - \alpha$.

(Convergence of r.v.) Name	Definition	written as	Example	
Convergence in Probability	If $\lim F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$.	$X_n \xrightarrow{d} X$ or $X_n \xrightarrow{d} F_X$	CLT 最弱:	见下面
Convergence in Distribution	If $\forall \epsilon > 0, \lim \mathbb{P}(X_n - X > \epsilon) = 0$.	$X_n \xrightarrow{p} X$	WLLN 中等:	If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$. $\Rightarrow \bar{X}_n \xrightarrow{p} \mu$
Almost Sure Convergence	If $\mathbb{P}(\lim X_n = X) = 1$	$X_n \xrightarrow{a.s.} X$	SLLN 最强:	If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$, $Var[X_i] < \infty. \Rightarrow \bar{X}_n \xrightarrow{a.s.} \mu$

Central Limit Theorem (CLT): If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$ and $Var[X_i] < \infty. \Rightarrow \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$ 样本均值的分布会趋近于正态分布

6.2 Test Statistics and Decision Rule

One Sample		Two Sample		
Find μ		Find μ		Find σ^2
Z-Test (σ^2 known) $Z = \frac{\bar{X} - \mu_0}{\sqrt{\sigma^2/n}} \sim N(0, 1)$	t-Test (σ^2 unknown) $T = \frac{\bar{X} - \mu_0}{\sqrt{S^2/n}} \sim t_{n-1}$	Paired t-Test (X,Y same Unit) $D_i = Y_i - X_i \sim N(\mu_D, \sigma_D^2)$ $T = \frac{\bar{D} - \mu_D}{\sqrt{S^2/n}} \sim t_{n-1}$	Two-Sample t-Test (独立) $T = \frac{(\bar{X} - \bar{Y}) - \Delta_0}{\sqrt{\sigma^2(\frac{1}{m} + \frac{1}{n})}} \sim N(0, 1)$ 未知: $S_p^2; T \sim t_{m+n-2}$	F-Test $F = \frac{S_X^2}{S_Y^2}$ $\sim F_{m-1, n-1}$

· 对于 Two-Sample t-Test 我们要求 X,Y σ^2 一致 ps: $H_0 : \mu_D = \mu_X - \mu_Y ; H_0 : \Delta_0 = \mu_X - \mu_Y ; X : m, Y : n$

Pooled Sample Variance: (Two sample Variance) $S_p^2 = \frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}$ ps:Unbiased

7 Writing Format

Distribution|Unbiased|Consistent

寻找新 **distribution**: 产生新 distribution 的一种方法: 通过条件概率来. $\mathbb{P}(A|B) = \mathbb{P}[(A \cap B)/B]$

Def|Unbiased:An estimator is unbiased if, on average, it equals the true value of the parameter it's estimating.

Def|Consistent: An estimator is consistent if it converges in probability to the true value of the parameter p as the sample size increases indefinitely.

MLE

Assumption|Calculate Estimator: Let random variables sequence $X_1, ..., X_n$ represent independent $\{? : e.g.sample\}$ from a common distribution $\{?\}$, represented by random variable X , that depends on the population parameter $\{?\}$.

Assumption|Calculate Maximum Likelihood: Suppose that we have n independent observations $x_1, ..., x_n$ drawn from an $\{?\}$ distribution, with probability density function $f_x = \{?\}$.

Reason| $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$: It's due to the invariant property of MLEs.

Reason|Newton's Method = (≠) Scoring Method: Because the second derivative of likelihood function doesn't (does) include the r.v.

Reason| $\mathbb{P}(\theta < ?) = \alpha$ is **false**: We cannot make such probability statement. $\{\theta\}$ is a fixed value (but unknown).

Geometric Comparison of $T_{Score}, T_{Wald}, T_{LR}$:

· **1. T_{Score}** (Slope Comparison): Represents the slope of the likelihood function at θ_0 ; A steeper slope indicates that the likelihood function is changing more rapidly at θ_0 , which implies a greater difference between $\hat{\theta}$ and θ_0 .

· **2. T_{Wald}** (Width Comparison): Compares the difference between $\hat{\theta}$ and θ_0 .

· **3. T_{LR}** (Height Comparison): Compares the MLE $L(\hat{\theta})$ and the likelihood value at $\theta_0, L(\theta_0)$. A larger height difference indicates a greater discrepancy between $\hat{\theta}$ and θ_0

Reason| $T_{Score}, T_{Wald}, T_{LR}$ **Other** 部分公式可以近似的理由: This is due to $\hat{\theta}$ being a consistent estimator of θ_0 , and $J(\theta_0)$ being a consistent estimator of $I(\theta_0)$

CI|PI

Def|CI: An interval of uncertainty for fitted regression line, given the explanatory variable x_0 .

Meaning|CI 表示含义: Not Reject: If data are generated under H_0 , there is a probability $(1 - \alpha)$ for the true θ_0 to be within the $(1 - \alpha)100\%$ CI. (i.e. If we were to repeat out experiment 100 times, we would expect the true θ_0 to be within the derived confidence interval (not necessarily CI:(a,b)) {e.g. 95} times.)

Assumption|CI: Let $X_1, ..., X_n$ be independent and identically distributed $\{? : distribution\}$ random variables, where $\{?\}$ is unknown/known ...

Reason| 误差来源: The source of uncertainty in (or in CI) an estimate comes from the principal of repeated sampling from the population.

Reason| 无误差:The true $\{?\}$ population parameter is assumed to be fixed (not random)

Def|PI: An interval of uncertainty for a future observation y_0 , given explanatory variable x_0 .

Reason|CI < PI: PI > CI, because it's accounts for the additional variability of the single observations, since $\sigma^2 > 0$

Reason| $\bar{x} = (left + right)/2$: The confidence interval for the $\{\mu\}$ is centered on $\bar{x}$

Hypothesis|P-value

p-value: p-value measures the plausibility that the data were generated under H_0 . **Low**

p-value: low p-value suggests H_0 is incompatible with the data and hence likely to be false.

Hypothesis Testing: Steps: ¹ State H_0, H_1 ; ² Select Test Statistic T ; ³ distribution of T under H_0 ; ⁴ observed test statistic; ⁵ critical value / region, p-value, or CI; ⁶ conclusion. || **Reject H_0** : We would reject H_0 in favour of the H_1 . (So that ...) || **Fail to reject H_0** : We fail to reject H_0 . Hence, there is strong evidence against the null hypothesis.

Bayes Inference

Meaning|Important Sampling (IS): It's a Monte Carlo technique that can be used when simulating from the posterior is not possible.

Meaning|Variation Methods: It minimizes a divergence between the posterior and the variation approximation.

Linear Regression

Explain|★Why considered to be linear: Since the responses are independent, have the same variance and the expected value of the response given covariates is linear in the unknown coefficients $\{? : e.g.\beta_0, \beta_1, \beta_2, ...\}$

Reason| 不说一个增加导致另一个...: Correlation is not Causation. A linear regression model can only describes correlated relationship.

Reason|MLE 和 LS 公式一样: The functional form of the (log-)likelihood being a sum of squared differences.

Assumption|Normal Linear Regression: Assumptions are that the observations: 1. are from a normal distribution; 2. are independent; 3. have a common variance; 4. Linearity of the expectation variable.

Meaning|LS 数值大小的表示含义: $S(\beta)$ small: μ_i relatively close to the y_i ; Large: μ_i far from their corresponding y_i .

Meaning| ε : The random component of the model is intended to capture the fact that if we gathered a replicate set of data, the β_i (coefficient) would not change, but the Y_i would, as a result of different measurement errors.

Reason|使用 **SL** 来算 β : Usually, Linear Model can be a system of n equations with p parameters with $n > p$. In this case, the system is overdetermined and there is no exact solution. The least square method is used to find the best solution.

Geometric Interpretation of Linear Model|LS Method: The least Square Estimation of Linear Model amounts to finding the *orthogonal projection* of the n dimensional response data **y** onto the p dimensional linear subspace spanned by the columns of **X**.

Geometric Interpretation of Linear Model| $\mathbb{E}[\mathbf{y}]$: $\mathbb{E}[\mathbf{y}]$ lies in the space spanned by the columns of **X**, | and the least square estimate seeks to find the point in this space that is closest to **y**.

Independent of $\hat{\beta}$, $\hat{\sigma}^2$: **f, r** independent $\Rightarrow \hat{\beta}, \hat{\sigma}^2$ independent.

When $Var(\varepsilon) = I_n \sigma^2$ (A = 0**)**: Implies that ε_i independent.

Reason| R^2, R^2_{adj} : **1**. The linear model explains $\{R^2\}$ of the Variability within the $\{y\}$ data. **2**. $\{1 - R\}$ of Variability of the $\{y\}$ data remains unexplained.

Reason| R^2, R^2_{adj} : **R** 越大 better predicts the data. || R^2 比较: As R^2 for the 'Model 1' is larger (smaller) than 'Model 2', so 'Model 1/2' is preferred as it explains more of the response variability.

Reason| 对于 **Factor** 类型的单个 **p-value** 不关心: If it's a model of Factors, it's often difficult to interpret the p-value for individual coefficients in any meaningful way. $\Rightarrow$ Prefer to test the whole factor effect.

Reliability of the prediction: 计算 $k = \frac{|x_0 - \bar{x}|}{S_x}$ ps: 一般来说小于 1.96 以上才说好的 reliability.

· Linear Regression 从 $\bar{x}$ 往左右, reliable 逐渐下降.

Reason|**y** 和 **x** 之间是否有任何关联: Association between the explanatory variable and the response is measured by the F statistic whose value is $\{?\}$ in this case. The corresponding p-value is small (large), so we do not reject (reject) the hypothesis of association at any reasonable significance level.

Reason| 是否能用边缘分布来推断 **(t-test)**: ¹ Case I: design matrix **X** orthogonal (or 其他方式得到 parameter independent), it's possible to use the marginal distribution of $\hat{\beta}$ to make inference. ² Case II: **X** not orthogonal (or 其他方式得到 parameter dependent), it's not possible to use the marginal distribution of $\hat{\beta}$ to make inference.

Method| 是否能用边缘分布来推断 **(t-test)**: 如果不能使用: Using 4.5 后半段内容.

***Not Exam|centered simple linear regression**: 将 x 变为 $x' = x - \bar{x}$, $\Rightarrow$ Design matrix will be orthogonal.

Method| 是否可以一次性 **drop** 多个参数 (β_i) : Tests are are valid for testing whether individual terms should be should be dropped from the model, but (Note that the MLEs of the tests are not independent, due to the design not being orthogonal) we cannot drop more than one term at a time on this basis.